United International **University** (**UIU**)
Dept. of Computer Science and Engineering (CSE)
Mid Term Assessment Year: **2024** Semester: **Summer**
Course: **CSE 323/CSE 3711/EEE 4413** Title: **Computer Networks**
Marks: **30** Time: **1 Hour 45 minutes**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

There are **3 (Three)** questions. Answer **all 3 (Three)** questions. All questions are of values indicated on the right-hand margin.

Q1. a) What is the difference between **communication** and **networking**? What are the key components of a data communication system? Explain with a diagram. **[1 + 2 = 3]**

b) Why is OSI model called a reference model? One of the layers of OSI reference model is responsible for moving frames from one hop (node) to the next. Which one is that? Draw a diagram to show the operational model of this layer from both the sender's and receiver's end and clearly label various items. **[1 + 1 + 1 = 3]**

c) For TCP/IP model, what is the application-specific address? Explain with a practical example. **[1 + 1 = 2]**

d) What is Daisy-chain topology and how does it differ from the Bus topology? Explain. **[2]**

e) What is Edge Router? Explain with a sample diagram. **[1 + 1 = 2]**

Q2. a) Suppose **10 cars** are traveling together as a caravan, following each other in a fixed order. Think of **each car** as a **bit** and **the caravan** as a **packet**. Also suppose that each toll booth services (that is, transmits) a car (1 bit in this case) at a rate of one car per **12 seconds**, and that it is late at night so that the caravan's cars are the only cars on the highway. A car can propagate at **100 km/hr**. If the setting is as in Fig. 1,

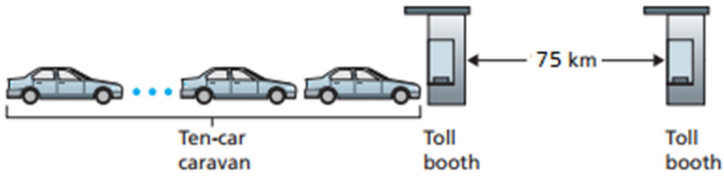


Fig. 1.

i. How long will it take for the caravan to be lined up before 2nd toll booth? **[2]**
ii. Will there be any car left at the first toll booth after 1 hour? **[1]**

b) In which layer of the OSI Reference Model a Router operates? Can a router run an application? Justify your answer. **[2]**

c) Is it possible to connect each access ISP with all other access ISPs in the Internet? Explain your answer. **[2]**

d) Describe with a clear diagram, the four sources of packet delay. **[3]**

Q3. a) Define **Transmission rate**, **Bandwidth**, and **Throughput**. What are the units used for each of these? **[3 + 1]**

b) What is web caching? Why is it done? **[2]**

c) See the table provided in Fig. 2. Fill in the blanks for underlying transport protocol (shown with '?' symbol) **[2]**

application	application layer protocol	underlying transport protocol
e-mail	SMTP [RFC 2821]	?
remote terminal access	Telnet [RFC 854]	?
Web	HTTP [RFC 2616]	?
file transfer	FTP [RFC 959]	?

Fig. 2.